Optimal Design of Linear Consecutive Systems*

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ABSTRACT

A critical issue for few nanometers technologies is the cost-yield balance, clearly tilted by soaring costs. An option to reduce costs, while also increasing yield, is to use reliability enhancement schemes. Unfortunately, these are considered power-hungry (due to redundancy), and entailing complex designs. From biology, neurons are prime examples of efficiency, achieving outstanding *communication reliabilities*, although relying on random ion channels. Aiming to bridge from biology to circuits, we will show how overlooked statistical results (about linear consecutive systems), combined with a Binet-like formula (for Fibonacci numbers of higher orders), allow avoiding lengthy reliability calculations, and present a straightforward neuron-inspired optimal design scheme for *reliable communications*.

CCS CONCEPTS

• Dependable and fault-tolerant systems and networks • Reliability • Network reliability • Hardware reliability

KEYWORDS

Consecutive system, Two-terminal network, Reliability

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1 Introduction

The latest chips use CMOS transistors in 3…5 nm processes, connected by a complex mesh of criss-crossing wires. Scaling requires new processing steps (EUV lithography), complicating fabrication, affecting yield, and hiking up prices. Yield is becoming an important constraint, hence *reliability* is of interest. Still, *designing for reliability* has not seen much progress, although efficient examples exists, like, e.g., the brain. Normally, explanations are on the lines of *"it is a different technology,"* but an understanding of how neurons are achieving outstanding reliability using redundancy, while still dissipating $10^{3…4}$ times less than the most advanced CMOS technologies, is still wanted.

Trying to understand neurons' *communications reliability* we will focus on *consecutive systems* and describe an optimization scheme entirely avoiding the lengthy/complex calculations of the reliability polynomials, or of their approximations or bounds.

Section 2 will go over the state-of-the-art of network reliability, mentioning consecutive systems in biology. Section 3 will present links between the reliability of linear consecutive systems at $p = 0.5$ and Fibonacci numbers of higher orders, and calculate these using a Binet-like formula. Section 4 will show how to accurately estimate the optimal k (for *communications*) if n is large, and present supporting simulations. Conclusions and further directions for research are going to end the paper.

2 State-of-the-Art

Network reliability was established in 1956 by Moore & Shannon [2] who abstracted a network N as a graph of nodes connected by edges, and assumed that the nodes always work, with only the edges failing independently with probability $(p = 1 - q)$. This led to a reliability polynomial $Rel(N; p)$.

Linear consecutive systems were introduced by Kontoleon in 1980 as "r-successive-out-of-n:F" [3], and rebranded "consecutive-k-out-of-n:F" ($N = Lin/Con/k/n$) by Chang & Niu [4]. Examples include street lights and gas/oil pipelines, while statistical results were published by de Moivre [5]. $Lin/Con/k/n$ are appealing as orderly spanning from series to parallel. Still, large n and optimal k require demanding calculations [6]. Fig. 1 presents $Lin/Con/3/16$, while corresponding simulations are detailed in Fig. 2(a), and optimal *communication* ($Rel > 90\%$) in Fig. 2(b).

$Lin/Con/k/n$ have not been used for circuits, probably as being wire-demanding (number and lengths), but are matching fluidics (oil/gas pipelines), and nano-fluidics in particular (Beiu & Dăuş in 2014 [7]). The regular organization observed by Xu et al. [8], revealed rings evenly spaced at 180-190 nm. Hence, axons are akin to nano-scaled oil/gas pipelines, equivalent to $Lin/Con/k/n$ with n of about 50,000 (1 cm / 190 nm) or higher, making exact calculations all but impossible [9].

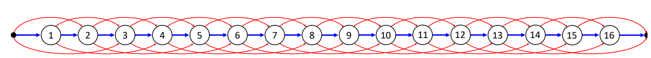

Figure 1: Linear consecutive-3-out-of-16:F.

* Extended version as [1]; expansion to 2D in progress [17].
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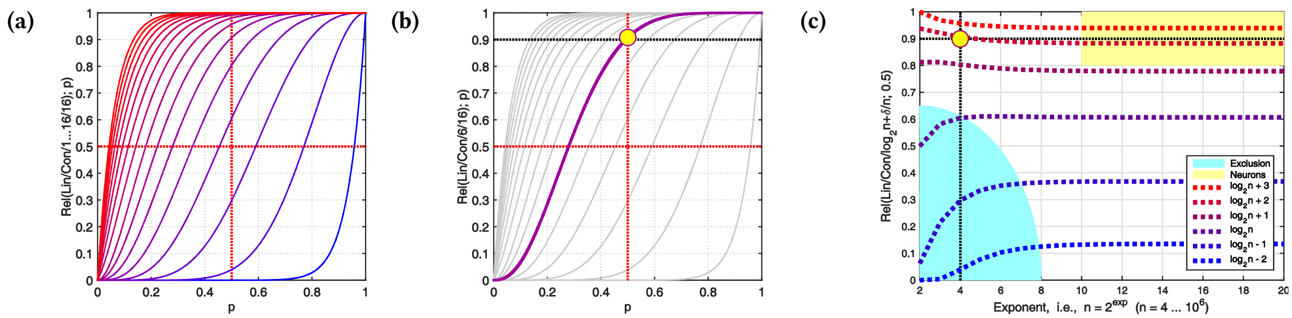

Figure 2: $Rel(Lin/Con/k/n; p)$: (a) $n = 16$ and $k = 1 \dots 16$; (b) optimal (communication) for $n = 16 = 2^4$ is achieved for $k = 6$ (i.e., $log_2 n + 2$); (c) exact simulation for $k = log_2 n + \delta$ (with $\delta = -2, -1, 0, 1, 2, 3$) and $n = 2^2 \dots 2^{20}$ (at $p = 0.5$).

3 Fibonacci and $Lin/Con/k/n$

Fibonacci numbers (A000045) were generalized by d'Ocagne [10] as $F_n^{(k)} = F_{n-1}^{(k)} + F_{n-2}^{(k)} + \cdots + F_{n-k}^{(k)}$, with the characteristic equation $x^k - x^{k-1} - \cdots - 1 = 0$ having the roots $\alpha_1, \dots, \alpha_k$.

Theorem 1 [11]**:** Let the positive root α_k of the characteristic equation be $2(1 - \varepsilon_k)$. Then $\varepsilon_k = \sum_{i \geq 1} \frac{1}{i} \binom{(k+1)i - 2}{i - 1} \frac{1}{2^{(k+1)i}}$. Hence α_k could be bounded as $2 - 2^{1-k} < \alpha_k < 2 - 2^{-k}$.

Theorem 2 [12]**:** A Fibonacci number of higher orders $F_n^{(k)}$ is $F_n^{(k)} = \text{rnd}\left[\frac{\alpha_k - 1}{2 + (k+1)(\alpha_k - 2)} \alpha_k^{n-1}\right]$ for all $n \geq 2 - k$, and for α_k the unique positive root of $x^k - x^{k-1} - \cdots - 1 = 0$.

Links to $Lin/Con/k/n$ (suggested by [13, 14]) established [15]:

$$Rel(Lin/Con/k/n; 0.5) = F_{n+2}^{(k)}/2^n. \tag{1}$$

4 Optimal Design for $Lin/Con/k/n$

Now we can estimate $Rel(Lin/Con/k/n; 0.5)$:

$$\frac{F_{n+2}^{(k)}}{2^n} \cong \frac{\frac{(\alpha_k - 1)}{2 + (k+1)(\alpha_k - 2)} \alpha_k^{n+1}}{2^n} = \frac{(\alpha_k - 1)\alpha_k}{2 - (2 - \alpha_k)(k+1)} \left(\frac{\alpha_k}{2}\right)^n = f(k) g(k, n) \tag{2}$$

by taking $\alpha_k \approx 2 - 2^{-k}$, setting $k = \log_2 n + \delta$, and calculating $\lim_{n \to \infty} g(\log_2 n + \delta, n) = e^{-1/2^{1+\delta}}$, as well as $\lim_{n \to \infty} f(\log_2 n + \delta)$ $= \lim_{n \to \infty} \frac{2^{1+\delta} n - 3}{2^{1+\delta} n - (\log_2 n + \delta + 1)} = 1$. Therefore:

$$\lim_{n \to \infty} Rel(Lin/Con/\log_2 n + \delta/n; 0.5) = e^{-1/2^{1+\delta}}. \tag{3}$$

Eq. (3) is based on approximations and limits, so *"How small can n be?"* We have computed $Rel(Lin/Con/\log_2 n + \delta/n; 0.5)$ exactly for $\delta = -2, -1, 0, 1, 2, 3$ and $n = 2^2 \dots 2^{20}$ (see Fig. 2(c)). For *communications* eq. (3) is valid for $n \geq 4$, and *the larger n the better*, suggesting that *communication reliability* on axons/dendrites could be easily estimated very quickly and accurately.

5 Conclusions

This short paper has introduced a scheme for $Lin/Con/k/n$ based on a Binet-like formula and unassuming approximations. Exact simulations have shown that this is valid for any $n \geq 4$. The optimal design of $Lin/Con/k/n$ for *communications* ($Rel > 90\%$) checks n versus 2^4 and selects k either $\log_2 n + 2$ or $\log_2 n + 3$ (see Fig. 2(c)). For $Lin/Con/k/n$ this is the simplest and fastest solution, while also being extremely precise for large n (checked up to $n = 2^{50}$). For axons $10^2 < n < 10^6$, but their cylindrical geometries translate into 2D (redundant) consecutive-(m, k)-out-of-(m, n) schemes (see [9, 16]). For these we have very recently determined that $k = \lceil(\log_2 n + 3)/m\rceil$ (to be detailed in [17]).

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